Does Batch Size Matter? Investigating the Impact of Batch Prompting on LTL-to-NL Translation Using Open-Source LLMs

I. Appendix

A. Prompting Techniques

Table I summarizes the seven prompting techniques employed throughout the batch prompting experiments, including the number of in-context examples used in each prompting configuration and whether an explanation is generated alongside the final natural language translation in the model output.

B. Prompt Templates

This section presents the complete prompt templates used to evaluate the effects of batch prompting across different models and prompting techniques for LTL-to-NL translation. Each prompt takes an LTL formula as input and instructs the model to generate its corresponding NL description. Depending on the prompting technique, some prompts additionally request explanations or step-by-step reasoning.

The prompts are adapted from prior work on LTL-to-NL translation. For all few-shot variants, a fixed set of seven in-context examples, drawn from the four datasets, was consistently used across prompting techniques. Reusing the same examples ensures a fair comparison by avoiding variability arising from example selection. The following subsections present the exact prompt templates for the seven prompting techniques (P1-P7).

1) P1 - Zero-shot: Transform the following LTL formula into a clear, precise natural-language specification, acting as a professional LTL expert.
Requirements: Preserve the original logical structure and all temporal / boolean constraints. Ensure that every atomic proposition and operator (`G`, `F`, `X`, `U`, `&`, `|`, `!`, $\rightarrow$) is faithfully reflected in the wording. Output only the English description; do not include any extra commentary, LTL syntax, or other formatting.

2) P2 - Zero-shot explanation-augmented: Translate the following LTL formula into a clear, concise natural-language sentence and explain your translation. Use only the operators `G`, `F`, `X`, `U`, `&`, `|`, `!`, `->`, and parentheses.
Output Format.

1) Natural Language Translation: `<English sentence here>`
2) Explanation: `<Explanation of how the LTL formula maps to the English description>`

3) P3 - Few-shot: Your task is to translate Linear Temporal Logic (LTL) formulas into clear, unambiguous English utterances. Use the examples below as guidance, then translate the input formula.
In-context Examples.

1) LTL: $!(\textit{green_room})\ U\ (\textit{landmark_2})$
   Utterance: Don't go into the green room until going to landmark two.
2) LTL: $F(\textit{green_room})\ \&\ G(!\textit{blue_room})$
   Utterance: Go to the green room but avoid the blue room.
3) LTL: $G(!\textit{Brook_St})\ \&\ F(\textit{Bookstore}\ \&\ F(\textit{Science_Library}))$
   Utterance: Do not enter Brook St; first find Bookstore and then find Science Library.
4) LTL: $G(\textit{JFcwdUeIa} \mid \textit{gdHweLFWxDqsFy})$
   Utterance: Always, JFcwdUeIa or gdHweLFWxDqsFy must hold.
5) LTL: $G((\textit{VjFeAl}\ \&\ \textit{pgCCaAqWFWIVqV}) \rightarrow F(\textit{fXWrgVJZjSEHK}\ \&\ \textit{oaDVhvA}))$
   Utterance: After both VjFeAl and pgCCaAqWFWIVqV, eventually both fXWrgVJZjSEHK and oaDVhvA occur.
6) LTL: $G(!(\textit{computer_starts}\ \&\ \textit{monitor_lights_up}\ \&\ \textit{cpu_is_cooled_down}))$
   Utterance: Absolutely never that the computer starts, the monitor lights up, and the CPU is cooled down.
7) LTL: $G((\textit{car_starts} \mid \textit{elevator_falls}) \rightarrow G(\textit{sensor_captures_data} \mid \textit{engine_breaks}))$
   Utterance: Whenever either the car starts or an elevator falls then, all the time a sensor captures data or the engine breaks.

4) P4 - Few-shot with reasoning: Translate the given LTL formula into a clear, concise English sentence. You are provided with worked examples below, each including step-by-step explanations. For the input formula, output only the final English translation; do not include any explanations.
In-context Examples

1) LTL: $!(\textit{green_room})\ U\ (\textit{landmark_2})$
   Explanation:
   - "*green_room*" means entering the green room.
   - "*!(green_room)*" means do not enter the green room.
   - "*landmark_2*" means reaching landmark two.
   - "*U*" means until.

   Final English: Don't go into the green room until going to landmark two.

TABLE I
PROMPTING TECHNIQUES EMPLOYED FOR BATCHED LTL-TO-NL TRANSLATION

Prompting techniques	Their descriptions	In-context examples	Explanation in output
Zero-shot (P1)	Uses only a task-level instruction describing the LTL-to-NL translation objective, requiring the model to directly generate the final NL output without in-context examples.	-	No
Zero-shot explanation-augmented (P2)	Extends the standard zero-shot setting by additionally requesting a brief textual explanation along with the generated NL output.	-	Yes
Few-shot (P3)	The model is guided with a LTL-to-NL pairs without any explanations.	7	No
Few-shot with reasoning (P4)	The model is provided with in-context examples that include step-by-step reasoning, while constraining the model to output only the final NL translation.	7	No
Few-shot explanation-augmented (P5)	The model is provided with in-context examples paired with explanatory descriptions and also instructs the model to generate both an explanation and the final NL output.	7	Yes
Zero-shot CoT (P6)	The model is explicitly instructed to reason step by step (e.g., using a prompt "Think through your translation step-by-step") before producing the final NL translation, without relying on in-context examples.	-	Yes
Few-shot explanatory CoT (P7)	Combines few-shot demonstrations with explicit chain-of-thought reasoning examples, guiding the model to generate a step-by-step explanation prior to the final NL description.	7	Yes

2) LTL: $F(\textit{green_room})$ & $G(!\textit{blue_room})$
Explanation:
   - "*green_room*" means going to the green room.
   - "$F(\textit{green_room})$" means eventually go to the green room.
   - "*blue_room*" means entering the blue room.
   - "!*blue_room*" means do not enter the blue room.
   - "$G(!\textit{blue_room})$" means always avoid the blue room.
   - "&" means and.

   Final English: Go to the green room but avoid the blue room.
3) LTL: $G(!\textit{Brook_St})$ & $F(\textit{Bookstore}$ & $F(\textit{Science_Library}))$
Explanation:
   - "!*Brook_St*" means do not enter Brook St.
   - "$G(!\textit{Brook_St})$" means always avoid Brook St.
   - "$F(\textit{Science_Library})$" means eventually find the Science Library.
   - "*Bookstore* & $F(\textit{Science_Library})$" means find the Bookstore and then eventually find the Science Library.
   - "$F(\cdot)$" means eventually.
   - "&" means both conditions must hold.

   Final English: Do not enter Brook St; first find Bookstore and then find Science Library.
4) LTL: $G(\textit{JFcwdUeIa} \mid \textit{gdHweLFWxDqsFy})$
Explanation:
   - "*JFcwdUeIa*" means JFcwdUeIa holds.
   - "*gdHweLFWxDqsFy*" means gdHweLFWxDqsFy holds.
   - "|" means either.
   - "$G(\cdot)$" means always.

   Final English: Always, JFcwdUeIa or gdHweLFWxDqsFy must hold.
5) LTL: $G((\textit{VjFeAl}$ & $\textit{pgCCaAqWFWIVqV}) \rightarrow F(\textit{fXWrgVJZjSEHK}$ & $\textit{oaDVhvA}))$
Explanation:
   - "*VjFeAl*" means VjFeAl occur.
   - "*VjFeAl*" and "*pgCCaAqWFWIVqV*" means both events occur.
   - "*fXWrgVJZjSEHK*" and "*oaDVhvA*" means fXWrgVJZjSEHK and oaDVhvA occur.
   - "$\rightarrow$" means if-then.
   - "$F(\cdot)$" means eventually.
   - "$G(\cdot)$" means always.

   Final English: After both VjFeAl and pgCCaAqWFWIVqV, eventually both fXWrgVJZjSEHK and oaDVhvA occur.
6) LTL: $G(!(\textit{computer_starts}$ & $\textit{monitor_lights_up}$ & $\textit{cpu_is_cooled_down}))$
Explanation:

- "computer_starts" means the computer starts.
- "monitor_lights_up" means the monitor lights up.
- "cpu_is_cooled_down" means the CPU is cooled down.
- "!" negates the conjunction.
- "$G(\cdot)$" means always.

Final English: Absolutely never that the computer starts, the monitor lights up, and the CPU is cooled down.

7) LTL: $G((\textit{car_starts} \mid \textit{elevator_falls}) \rightarrow G(\textit{sensor_captures_data} \mid \textit{engine_breaks}))$
Explanation:
   - "$\textit{car_starts} \mid \textit{elevator_falls}$" means either the car starts or an elevator falls.
   - "$\textit{sensor_captures_data} \mid \textit{engine_breaks}$" means either a sensor captures data or the engine breaks.
   - "$G(\textit{sensor_captures_data} \mid \textit{engine_breaks})$" means always, either a sensor captures data or the engine breaks.
   - "|" denotes logical disjunction (either).
   - "$\rightarrow$" denotes logical implication.
   - "$G(\cdot)$" denotes the globally (always) temporal operator.

   Final English: Whenever either the car starts or an elevator falls then, all the time a sensor captures data or the engine breaks.

5) P5 - Few-shot explanation-augmented: You are a Linear Temporal Logic (LTL) expert. Translate the following LTL formula into concise, precise natural-language (NL) sentences and explain your translation step by step. Remember that X means "next", U means "until", G means "globally" (always), and F means "finally" (eventually). Parentheses specify operator precedence and group subformulas together. You must answer in the following format: [

Explanation: `<Step-by-step reasoning>`

So the Final NL: `<Final natural-language translation>`

]

In-context Examples.

1) LTL: $!(\textit{green_room})\ U\ (\textit{landmark_2})$
Explanation:
   - "*green_room*" means entering the green room.
   - "$!(\textit{green_room})$" means do not enter the green room.
   - "*landmark_2*" means reaching landmark two.
   - "U" means until.

   Final NL: Don't go into the green room until going to landmark two.
   FINISH
2) LTL: $F(\textit{green_room}) \ \&\ G(!\textit{blue_room})$
Explanation:
   - "*green_room*" means going to the green room.
   - "$F(\textit{green_room})$" means eventually go to the green room.
   - "*blue_room*" means entering the blue room.
   - "$!\textit{blue_room}$" means do not enter the blue room.
   - "$G(!\textit{blue_room})$" means always avoid the blue room.
   - "&" means and.

   Final NL: Go to the green room but avoid the blue room.
   FINISH
3) LTL: $G(!\textit{Brook_St}) \ \&\ F(\textit{Bookstore} \ \&\ F(\textit{Science_Library}))$
Explanation:
   - "$!\textit{Brook_St}$" means do not enter Brook St.
   - "$G(!\textit{Brook_St})$" means always avoid Brook St.
   - "$F(\textit{Science_Library})$" means eventually find the Science Library.
   - "$\textit{Bookstore} \ \&\ F(\textit{Science_Library})$" means find the Bookstore and then eventually find the Science Library.
   - "$F(\cdot)$" means eventually.
   - "&" means both conditions must hold.

   Final NL: Do not enter Brook St; first find Bookstore and then find Science Library.
   FINISH
4) LTL: $G(\textit{JFcwdUeIa} \mid \textit{gdHweLFWxDqsFy})$
Explanation:
   - "*JFcwdUeIa*" means JFcwdUeIa holds.
   - "*gdHweLFWxDqsFy*" means gdHweLFWxDqsFy holds.
   - "|" means either.
   - "$G(\cdot)$" means always.

   Final NL: Always, JFcwdUeIa or gdHweLFWxDqsFy must hold.
   FINISH
5) LTL: $G((\textit{VjFeAl} \ \&\ \textit{pgCCaAqWFWIVqV}) \rightarrow F(\textit{fXWrgVJZjSEHK} \ \&\ \textit{oaDVhvA}))$
Explanation:
   - "*VjFeAl*" means VjFeAl occur.
   - "*VjFeAl*" and "*pgCCaAqWFWIVqV*" means both events occur.
   - "*fXWrgVJZjSEHK*" and "*oaDVhvA*" means fXWrgVJZjSEHK and oaDVhvA occur.
   - "$\textit{(VjFeAl \& pgCCaAqWFWIVqV)} \rightarrow F(...)$" means if both VjFeAl and pgCCaAqWFWIVqV happen, then eventually both fXWrgVJZjSEHK and oaDVhvA happen.
   - "$\rightarrow$" means if-then.
   - "$F(\cdot)$" means eventually.
   - "$G(\cdot)$" means always.

   Final NL: After both VjFeAl and pgCCaAqWFWIVqV, eventually both fXWrgVJZjSEHK and oaDVhvA occur.
   FINISH
6) LTL: $G(!(\textit{computer_starts} \ \&\ \textit{monitor_lights_up} \ \&\ \textit{cpu_is_cooled_down}))$
Explanation:
   - "computer_starts" means the computer starts.
   - "monitor_lights_up" means the monitor lights up.
   - "cpu_is_cooled_down" means the CPU is cooled down.

- The conjunction represents all three events occurring together.
- "!" negates the conjunction.
- "$G(\cdot)$" means always.

Final NL: Absolutely never that the computer starts, the monitor lights up, and the CPU is cooled down.
FINISH

7) LTL: $G((\textit{car_starts} \mid \textit{elevator_falls}) \rightarrow G(\textit{sensor_captures_data} \mid \textit{engine_breaks}))$
Explanation:
   - "$\textit{car_starts} \mid \textit{elevator_falls}$" means either the car starts or an elevator falls.
   - "$\textit{sensor_captures_data} \mid \textit{engine_breaks}$" means either a sensor captures data or the engine breaks.
   - "$G(\textit{sensor_captures_data} \mid \textit{engine_breaks})$" means always, either a sensor captures data or the engine breaks.
   - "|" means either.
   - "$\rightarrow$" means implication.
   - "$G(\cdot)$" means always.

   Final NL: Whenever either the car starts or an elevator falls then, all the time a sensor captures data or the engine breaks.
   FINISH

6) P6 - Zero-shot CoT: You are a Linear Temporal Logic (LTL) expert. For each formula I give to you, "Think through the translation step by step", then output a concise natural-language instruction. Remember that X means "next", U means "until", G means "globally", F means "finally", and GF means "infinitely often". Parentheses specify operator precedence and group subformulas together.
Must answer in the following format: [
Explanation: `<Give explanation here>`
So the Final NL: `<Final natural-language translation>`
]

7) P7 - Few-shot explanatory CoT: You are a Linear Temporal Logic (LTL) expert. Your answers must follow the output format below as indicated in the examples and adhere to the following rules:

1) Remember that U means "until", G means "globally / always", F means "finally / eventually", and GF means "infinitely often".
2) Only the following operators may appear: |, &, !, $\rightarrow$.
3) Atomic propositions must be interpreted consistently across the explanation and the final instruction.

Translate the following LTL formula into clear, step by step natural-language instructions. Show your chain-of-thought briefly, then provide the final NL instruction.

1) LTL formula: $!(\textit{green_room})\ U\ (\textit{landmark_2})$
Explanation dictionary:
   - "$!(\textit{green_room})\ U\ (\textit{landmark_2})$" : avoid the green room until landmark two is reached
   - "*green_room*" : green room
   - "*landmark_2*" : landmark two

   Final natural-language instruction: "Don't go into the green room until going to landmark two." FINISH
2) LTL formula: $F(\textit{green_room})\ \&\ G(!\textit{blue_room})$
Explanation dictionary:
   - "$F(\textit{green_room})\ \&\ G(!\textit{blue_room})$" : eventually reach the green room while always avoiding the blue room
   - "*green_room*" : green room
   - "*blue_room*" : blue room

   Final natural-language instruction: "Go to the green room but avoid the blue room." FINISH
3) LTL formula: $G(!\textit{Brook_St})\ \&\ F(\textit{Bookstore}\ \&\ F(\textit{Science_Library}))$
Explanation dictionary:
   - "$G(!\textit{Brook_St})$" : always avoid Brook St
   - "$F(\textit{Bookstore}\ \&\ F(\textit{Science_Library}))$" : first go to the Bookstore, then eventually reach the Science Library

   Final natural-language instruction: "Do not enter Brook St; first find Bookstore and then find Science Library." FINISH
4) LTL formula: $G(\textit{JFcwdUeIa} \mid \textit{gdHweLFWxDqsFy})$
Explanation dictionary:
   - "$G(\textit{JFcwdUeIa} \mid \textit{gdHweLFWxDqsFy})$" : always either JFcwdUeIa or gdHweLFWxDqsFy must hold
   - "*JFcwdUeIa*" : JFcwdUeIa
   - "*gdHweLFWxDqsFy*" : gdHweLFWxDqsFy

   Final natural-language instruction: "Always, JFcwdUeIa or gdHweLFWxDqsFy must hold." FINISH
5) LTL formula: $G((\textit{VjFeAl}\ \&\ \textit{pgCCaAqWFWIVqV}) \rightarrow F(\textit{fXWrgVJZjSEHK}\ \&\ \textit{oaDVhvA}))$
Explanation dictionary:
   - "$(\textit{VjFeAl}\ \&\ \textit{pgCCaAqWFWIVqV})$" : both VjFeAl and pgCCaAqWFWIVqV occur
   - "$F(\textit{fXWrgVJZjSEHK}\ \&\ \textit{oaDVhvA})$" : eventually both fXWrgVJZjSEHK and oaDVhvA occur
   - "$G((\cdot) \rightarrow (\cdot))$" : always, if the condition holds, then the consequence eventually holds

   Final natural-language instruction: "After both VjFeAl and pgCCaAqWFWIVqV, eventually both fXWrgVJZjSEHK and oaDVhvA occur." FINISH
6) LTL formula: $G(!(\textit{computer_starts}\ \&\ \textit{monitor_lights_up}\ \&\ \textit{cpu_is_cooled_down}))$
Explanation dictionary:
   - "$\textit{computer_starts}\ \&\ \textit{monitor_lights_up}\ \&\ \textit{cpu_is_cooled_down}$" : all three events occur simultaneously
   - "$!(\cdot)$" : not
   - "$G(!(\cdot))$" : always never allow this to happen

   Final natural-language instruction: "Absolutely never that the computer starts, the monitor lights up, and the CPU is cooled down." FINISH
7) LTL formula: $G((\textit{car_starts} \mid \textit{elevator_falls}) \rightarrow G(\textit{sensor_captures_data} \mid \textit{engine_breaks}))$
Explanation dictionary:

- "*car_starts* | *elevator_falls*" : either the car starts or an elevator falls
- "G(*sensor_captures_data* | *engine_breaks*)" : always either a sensor captures data or the engine breaks
- "$G((\cdot) \rightarrow G(\cdot))$" : always, if the condition occurs, then the consequence always holds

Final natural-language instruction: "Whenever either the car starts or an elevator falls then, all the time a sensor captures data or the engine breaks." FINISH

C. Detailed Natural Language Evaluation Results Across Batch Sizes

Tables II, III, IV, and V present the complete natural-language evaluation results for batch sizes 2, 4, 8, and 16, respectively. The tables report metric-wise translation scores across all 10 evaluated models and 7 prompting strategies, including lexical, syntactic, semantic, and average total scores under each batch sizes.

TABLE II
NATURAL LANGUAGE EVALUATION ACROSS MODEL CATEGORIES FOR BATCH SIZE 2

Category	Model	Prompt	Natural Language Evaluation						
			Lexical			Syntactic	Semantic	Total Score (TS)	Avg TS
			BLEU	ROUGE-L	METEOR	TSED	SBERT		
Tiny Models	Falcon-H1-1.5B-Deep-Instruct	P1	21.34	39.19	38.97	0.86	76.89	48.57	54.21
		P2	22.90	40.31	44.06	0.54	76.19	48.93	
		P3	36.13	58.32	57.56	10.50	87.97	61.29	
		P4	32.25	54.22	54.54	3.14	86.69	58.07	
		P5	29.46	51.11	54.48	1.20	86.43	56.96	
		P6	22.79	41.09	45.42	0.46	81.55	51.80	
		P7	28.52	46.41	51.38	0.95	82.05	53.85	
	Gemma-2-2B-it	P1	25.44	44.57	38.79	0.29	73.08	47.48	51.33
		P2	22.97	38.86	42.59	0.90	70.17	45.71	
		P3	34.22	55.93	55.64	5.87	86.08	58.79	
		P4	19.43	44.79	35.29	0.55	75.23	47.68	
		P5	33.13	53.44	53.54	4.25	85.70	57.71	
		P6	24.78	42.31	35.82	2.90	72.44	47.09	
		P7	29.45	51.40	51.96	1.54	82.58	54.88	
Small Models	Falcon-H1-7B-Instruct	P1	26.63	42.99	55.32	0	70.49	47.74	54.92
		P2	26.37	44.62	50.36	0.80	75.21	49.90	
		P3	31.25	53.20	53.12	2.91	85.73	57.20	
		P4	35.54	55.87	57.83	6.03	86.47	59.37	
		P5	36.15	54.48	56.80	4.91	82.77	57.11	
		P6	29.63	49.71	50.64	1.14	83.87	55.16	
		P7	32.60	53.95	56.72	2.06	86.39	57.93	
	MetaLlama-3-8B-Instruct	P1	23.13	37.01	43.14	0.08	72.66	46.67	53.96
		P2	23.31	34.02	41.93	1	65.67	42.96	
		P3	41.66	62.91	62.99	8.95	88.99	63.04	
		P4	38.92	60.99	61.06	9.52	87.17	61.59	
		P5	37.54	58.66	58.60	7.01	87.74	60.75	
		P6	24.67	38.18	45.25	0.98	70.76	46.39	
		P7	34.22	53.43	55.32	6.36	81.56	56.35	
Medium Models	Mistral-Small-3.2-24B-Instruct	P1	22.90	36.72	46.61	0	76.22	48.73	55.95
		P2	29.53	44.45	50.98	1.94	75.98	50.87	
		P3	39.16	57.24	61.69	6.18	87.01	60.55	
		P4	36.36	59.60	60.44	5.59	88.13	60.82	
		P5	34.91	57.98	59.40	6.59	87.70	60.40	
		P6	33.02	48.77	51.60	3.24	80.23	54.10	
		P7	32.71	54.26	58.55	2.11	82.42	56.18	
	Gemma-3-27b-it	P1	29.45	47.13	50.01	2.60	76.39	51.37	54.30
		P2	24.91	39.36	43.01	1.64	69.03	45.57	
		P3	47.66	65.84	66.14	15.59	90.32	66.24	
		P4	39.36	62.30	61.72	9.90	89.27	62.95	
		P5	34.74	57.10	56.58	5.49	87.74	59.81	
		P6	24.43	44.11	41.37	1.69	79.20	50.93	
		P7	30.90	48.51	54.62	0	59.70	43.25	
Large Models	Qwen2.5-72B-Instruct	P1	28.62	45.58	51.64	1.78	77.08	51.48	56.21
		P2	25.76	45.12	49.12	1.84	78.52	51.63	
		P3	39.42	60.60	61.60	8.49	89.79	62.76	
		P4	37.49	61.40	61.05	8.84	89.31	62.42	
		P5	33.45	54.88	54.73	3.96	86.01	58.10	
		P6	22.88	41.47	43.22	1.04	75.46	48.70	
		P7	34.52	55.40	59.74	4.47	85.08	58.40	
	Llama-2-70B-chat	P1	24.61	37.33	40.80	1.08	69.99	45.49	51.20
		P2	24.15	37.48	42.54	0.56	67.59	44.32	
		P3	33.94	58.04	57.41	6.86	86.08	59.35	
		P4	30.06	48.09	48.97	3.77	76.09	51.51	
		P5	31.78	49.87	50.40	3.22	77.58	52.64	
		P6	23.26	36.83	41.13	0	71.88	46.06	
		P7	34.69	57.24	58.77	4.63	86.08	59.04	
Ultra-Large Models	Mixtral-8x22B-Instruct-v0.1	P1	26.58	42.67	50.06	1.18	76.29	50.31	50.55
		P2	21.53	39.69	42.63	0.08	75.36	48.08	
		P3	38.88	60.71	61.38	10.34	88.14	62.24	
		P4	32.12	55.24	55.46	5.76	85.22	58.04	
		P5	12.18	22.08	25.34	1.01	47.79	30.06	
		P6	26.40	45.68	48.08	0.66	79.26	51.78	
		P7	27.09	45.52	49.73	1.76	81.55	53.36	
	Llama-4-Scout-17B-16E-Instruct 109B	P1	30.15	44.10	53.35	0.27	78.19	51.91	56.39
		P2	26.87	39.72	47.36	2.24	74.94	49.31	
		P3	42.12	62.63	61.94	16.49	88.75	64.34	
		P4	37.02	59.72	60.28	8.64	87.25	61.06	
		P5	36.32	58.12	58.54	7.54	87.86	60.74	
		P6	27.20	43.64	46.97	1.24	74.61	49.33	
		P7	34.41	54.59	58.06	2.42	85.65	58.02	

TABLE III
NATURAL LANGUAGE EVALUATION ACROSS MODEL CATEGORIES FOR BATCH SIZE 4

Category	Model	Prompt	Natural Language Evaluation						
			Lexical			Syntactic	Semantic	Total Score (TS)	Avg TS
			BLEU	ROUGE-L	METEOR	TSED	SBERT		
Tiny Models	Falcon-H1-1.5B-Deep-Instruct	P1	13.71	39.73	25.91	0.00	74.40	45.14	53.33
		P2	26.16	44.47	48.16	0.29	81.64	52.76	
		P3	31.14	51.48	52.35	1.38	86.13	56.84	
		P4	28.13	52.70	48.57	1.86	85.64	56.13	
		P5	30.07	52.78	52.80	1.00	85.86	56.70	
		P6	23.09	44.33	40.07	0.69	79.40	50.59	
		P7	27.58	50.63	51.73	1.65	83.67	55.15	
	Gemma-2-2B-it	P1	25.09	46.70	42.99	1.03	78.22	50.79	51.32
		P2	17.37	39.68	29.51	0.00	70.75	44.03	
		P3	32.23	52.49	53.55	7.58	84.34	57.51	
		P4	21.27	51.58	38.04	2.19	81.24	52.14	
		P5	26.68	55.16	51.18	2.91	84.59	56.17	
		P6	15.86	41.22	30.12	0.16	67.72	42.61	
		P7	30.58	51.83	53.79	1.47	84.20	56.01	
Small Models	Falcon-H1-7B-Instruct	P1	18.12	39.84	37.47	0.00	67.73	43.41	54.39
		P2	26.62	47.00	50.40	1.16	73.83	49.55	
		P3	35.90	56.50	57.69	5.69	86.32	59.31	
		P4	32.80	56.57	50.39	5.40	86.04	58.08	
		P5	33.77	54.01	54.98	5.11	83.39	56.99	
		P6	30.17	50.95	47.88	1.65	82.79	54.63	
		P7	34.05	54.57	59.92	1.76	87.13	58.77	
	MetaLlama-3-8B-Instruct	P1	24.89	38.91	45.24	1.26	72.11	47.21	54.61
		P2	24.17	34.91	44.17	0.89	66.93	43.97	
		P3	40.24	62.43	62.39	6.86	87.91	61.83	
		P4	37.09	60.56	61.00	7.84	87.84	61.35	
		P5	34.33	58.16	57.78	5.91	86.19	59.30	
		P6	26.32	44.16	46.94	2.20	74.60	49.48	
		P7	35.87	58.01	57.60	3.85	86.37	59.10	
Medium Models	Mistral-Small-3.2-24B-Instruct	P1	27.28	40.99	53.32	0.01	64.80	44.56	55.43
		P2	30.21	47.13	51.65	3.39	75.54	51.35	
		P3	44.67	61.89	62.80	8.11	89.16	63.14	
		P4	37.06	59.87	59.89	5.75	88.81	61.24	
		P5	35.21	58.62	59.12	6.63	85.83	59.54	
		P6	26.43	43.44	45.61	0.18	77.38	50.27	
		P7	33.11	54.42	58.69	2.70	85.47	57.90	
	Gemma-3-27b-it	P1	29.01	46.64	50.48	2.63	75.96	51.12	54.77
		P2	25.92	41.23	45.15	2.41	72.50	47.96	
		P3	47.55	65.02	65.97	12.01	90.15	65.33	
		P4	36.24	60.37	60.80	7.17	89.18	61.77	
		P5	34.74	59.15	59.38	5.49	87.34	60.10	
		P6	26.41	45.23	43.37	1.69	81.37	52.52	
		P7	32.12	49.47	55.17	0.94	61.49	44.61	
Large Models	Qwen2.5-72B-Instruct	P1	28.66	46.42	50.57	1.35	82.47	54.07	57.02
		P2	25.76	45.12	49.12	1.84	78.41	51.57	
		P3	38.53	59.77	61.04	8.91	94.03	64.73	
		P4	37.64	62.33	61.99	9.87	89.40	62.87	
		P5	32.66	54.44	55.45	3.50	86.41	58.16	
		P6	22.27	41.12	42.84	0.58	85.07	53.27	
		P7	31.68	50.98	55.41	3.89	79.74	54.46	
	Llama-2-70B-chat	P1	26.42	41.22	44.37	1.46	71.49	47.24	49.75
		P2	24.52	38.24	42.39	0.35	67.66	44.42	
		P3	34.59	60.09	59.28	6.71	88.23	60.85	
		P4	23.70	38.39	39.17	3.00	65.37	43.41	
		P5	28.93	45.12	46.11	3.03	69.41	47.33	
		P6	22.80	39.11	42.79	0.09	72.57	46.77	
		P7	31.32	56.15	55.90	5.43	85.64	58.24	
Ultra-Large Models	Mixtral-8x22B-Instruct-v0.1	P1	28.03	44.78	52.01	1.40	77.91	51.72	51.99
		P2	21.57	38.14	40.31	0.07	71.55	45.79	
		P3	38.25	62.39	62.34	10.05	89.17	62.89	
		P4	35.34	57.37	58.85	5.36	84.25	58.35	
		P5	17.20	29.22	33.28	0.14	59.88	37.94	
		P6	27.25	46.62	50.14	0.93	81.05	53.11	
		P7	27.73	46.06	50.52	3.85	81.85	54.13	
	Llama-4-Scout-17B-16E-Instruct 109B	P1	30.05	44.05	52.56	1.08	79.23	52.50	57.43
		P2	28.19	42.92	50.27	1.84	76.59	50.80	
		P3	43.66	63.54	63.28	17.91	89.54	65.40	
		P4	37.62	60.83	61.53	8.39	88.80	62.08	
		P5	35.94	59.58	59.43	7.34	87.99	60.96	
		P6	27.03	46.02	50.04	1.10	78.85	51.95	
		P7	34.01	55.90	58.46	3.29	85.63	58.31	

TABLE IV
NATURAL LANGUAGE EVALUATION ACROSS MODEL CATEGORIES FOR BATCH SIZE 8

Category	Model	Prompt	Natural Language Evaluation						
			Lexical			Syntactic	Semantic	Total Score (TS)	Avg TS
			BLEU	ROUGE-L	METEOR	TSED	SBERT		
Tiny Models	Falcon-H1-1.5B-Deep-Instruct	P1	20.90	41.98	37.26	0.74	77.66	48.99	53.53
		P2	25.01	46.29	45.09	0.90	79.05	51.34	
		P3	30.49	54.25	52.95	2.31	85.95	57.21	
		P4	31.55	53.18	52.94	3.96	85.57	57.34	
		P5	32.51	54.41	56.10	2.19	86.88	58.18	
		P6	23.50	37.27	46.49	0.00	71.39	46.42	
		P7	29.37	51.69	50.46	1.74	83.38	55.19	
	Gemma-2-2B-it	P1	27.54	47.67	44.70	0.28	77.76	50.93	53.10
		P2	26.45	46.48	43.90	2.31	77.27	50.78	
		P3	37.03	55.81	57.18	8.93	84.91	59.24	
		P4	23.34	50.13	45.14	1.78	81.88	53.16	
		P5	28.61	54.26	50.75	2.26	86.83	57.23	
		P6	20.93	42.03	36.79	0.00	72.83	46.39	
		P7	28.78	49.09	49.27	0.78	82.20	53.97	
Small Models	Falcon-H1-7B-Instruct	P1	25.04	43.53	45.29	0.00	73.97	48.37	55.02
		P2	26.23	44.70	50.02	2.12	78.29	51.66	
		P3	33.74	57.14	53.91	6.86	87.23	59.47	
		P4	35.43	54.96	57.78	4.45	86.00	58.71	
		P5	34.64	55.48	56.34	4.61	86.80	58.97	
		P6	26.55	45.95	42.05	1.78	75.09	49.36	
		P7	34.12	55.11	59.40	3.12	86.28	58.63	
	MetaLlama-3-8B-Instruct	P1	24.07	36.97	43.26	1.88	69.25	45.43	55.16
		P2	25.18	39.79	47.37	1.23	76.78	49.87	
		P3	38.36	61.77	61.44	5.85	89.13	61.89	
		P4	36.32	59.61	59.57	5.04	88.82	60.97	
		P5	33.56	57.94	55.72	6.80	86.53	59.35	
		P6	23.17	42.09	43.28	0.87	74.38	48.22	
		P7	39.02	59.52	58.74	6.45	86.81	60.42	
Medium Models	Mistral-Small-3.2-24B Instruct	P1	17.58	37.72	35.66	0.00	73.97	46.08	57.40
		P2	27.16	44.79	51.00	0.79	86.34	55.62	
		P3	38.08	59.03	60.07	4.58	87.83	60.55	
		P4	38.20	60.25	60.83	6.66	88.76	61.64	
		P5	36.96	58.44	60.44	5.98	87.89	60.73	
		P6	31.88	52.44	54.73	3.12	86.56	57.81	
		P7	32.50	51.30	56.31	2.55	89.76	59.40	
	Gemma-3-27b-it	P1	28.66	47.81	50.09	1.71	77.52	51.76	53.66
		P2	18.12	27.91	30.34	1.38	49.34	32.58	
		P3	47.79	65.73	65.17	12.23	91.22	65.93	
		P4	38.04	60.91	60.88	11.29	89.38	62.93	
		P5	34.76	59.10	56.58	5.36	91.74	61.99	
		P6	24.43	44.11	41.37	1.56	79.20	50.90	
		P7	36.36	55.63	62.20	1.60	67.59	49.53	
Large Models	Qwen2.5-72B-Instruct	P1	28.54	47.45	50.73	1.71	79.21	51.15	53.31
		P2	27.09	45.12	52.02	1.84	78.72	52.15	
		P3	40.01	60.06	62.47	7.62	89.89	62.72	
		P4	37.42	61.35	61.33	6.64	88.95	61.81	
		P5	32.66	54.44	55.45	3.50	86.41	58.16	
		P6	22.52	41.12	43.20	0.58	75.12	48.36	
		P7	23.06	35.04	39.14	3.20	56.91	38.82	
	Llama-2-70B-chat	P1	27.26	42.07	45.07	2.91	67.11	45.58	52.01
		P2	22.86	38.54	42.26	0.24	65.84	43.33	
		P3	32.12	58.22	56.09	5.55	87.34	59.42	
		P4	31.52	53.38	54.52	5.07	80.66	55.29	
		P5	28.41	51.57	50.27	1.05	85.80	56.14	
		P6	21.60	41.70	43.48	0.07	73.30	47.34	
		P7	31.14	54.02	54.98	3.65	84.46	56.97	
Ultra-Large Models	Mixtral-8x22B-Instruct-v0.1	P1	29.83	48.21	54.63	1.10	80.14	53.56	51.99
		P2	20.20	35.71	39.22	0.61	74.41	46.84	
		P3	39.30	62.35	62.22	8.82	89.36	62.83	
		P4	37.05	59.64	60.34	7.52	89.02	61.72	
		P5	17.20	29.22	33.28	0.14	59.88	37.94	
		P6	27.96	48.55	50.54	0.98	80.96	53.38	
		P7	23.02	37.79	41.49	2.14	73.99	47.65	
	Llama-4-Scout-17B-16E-Instruct 109B	P1	29.18	45.28	54.37	0.93	79.60	52.87	53.15
		P2	18.74	29.78	36.90	0.97	54.94	36.21	
		P3	40.75	62.90	62.01	14.69	89.63	64.32	
		P4	37.10	59.75	60.32	8.72	88.59	61.76	
		P5	32.31	51.74	52.81	6.50	78.14	54.06	
		P6	22.89	41.51	45.22	1.17	72.96	47.68	
		P7	33.82	52.34	54.93	4.78	80.19	55.16	

TABLE V
NATURAL LANGUAGE EVALUATION ACROSS MODEL CATEGORIES FOR BATCH SIZE 16

Category	Model	Prompt	Natural Language Evaluation						
			Lexical			Syntactic	Semantic	Total Score (TS)	Avg TS
			BLEU	ROUGE-L	METEOR	TSED	SBERT		
Tiny Models	Falcon-H1-1.5B-Deep-Instruct	P1	17.58	44.30	33.26	0.63	76.05	47.67	52.06
		P2	23.89	51.34	47.08	1.67	84.19	54.66	
		P3	24.31	51.31	45.48	0.89	83.53	54.05	
		P4	20.77	53.02	39.69	0.63	84.84	53.89	
		P5	27.27	45.68	48.84	1.47	79.73	52.34	
		P6	21.77	41.45	45.01	0.16	82.61	52.16	
		P7	21.11	46.70	40.34	0.30	77.49	49.62	
	Gemma-2-2B-it	P1	28.33	48.09	45.23	1.99	78.66	51.89	49.99
		P2	17.67	38.40	33.74	0.89	71.30	44.81	
		P3	25.16	52.77	41.97	6.13	82.38	54.41	
		P4	25.28	47.29	47.20	1.95	81.18	52.96	
		P5	27.11	51.36	47.88	2.34	79.08	52.64	
		P6	12.12	38.86	21.20	0.00	61.30	37.87	
		P7	29.99	51.59	47.27	3.58	83.48	55.34	
Small Models	Falcon-H1-7B-Instruct	P1	31.20	56.26	50.47	5.64	85.90	57.87	55.25
		P2	27.73	46.04	49.14	3.09	74.94	50.38	
		P3	30.45	56.72	53.53	5.47	87.37	58.85	
		P4	30.94	54.53	52.03	3.81	85.38	57.20	
		P5	27.87	46.40	51.47	2.42	79.68	52.90	
		P6	24.97	53.07	38.26	2.21	83.17	53.66	
		P7	32.06	53.00	53.94	1.15	83.46	55.86	
	MetaLlama-3-8B-Instruct	P1	21.90	39.70	41.66	2.56	75.97	48.82	55.32
		P2	21.08	39.54	41.62	0.68	74.88	47.80	
		P3	38.36	61.77	61.44	5.85	89.13	61.89	
		P4	35.35	58.56	59.82	7.33	89.02	61.35	
		P5	33.51	57.62	57.57	4.12	86.58	58.98	
		P6	21.84	38.11	38.36	0.71	74.90	47.42	
		P7	38.22	59.91	58.31	11.21	86.12	60.95	
Medium Models	Mistral-Small-3.2-24-	P1	18.33	40.40	36.20	0.00	71.67	45.33	54.88
		P2	27.91	46.13	52.91	1.38	79.18	52.56	
		P3	41.67	61.94	62.43	10.46	88.43	62.91	
		P4	35.42	59.58	54.65	5.94	87.98	60.14	
		P5	31.30	51.95	54.41	4.25	83.33	56.28	
		P6	25.46	46.78	46.39	1.89	82.72	53.60	
		P7	31.75	48.88	54.55	1.95	78.92	53.37	
	Gemma-3-27b-it	P1	26.08	47.73	46.17	1.70	77.23	50.95	56.31
		P2	34.85	47.99	52.41	3.89	80.65	54.63	
		P3	43.89	64.51	62.08	8.51	90.23	63.87	
		P4	38.08	60.27	61.69	10.01	89.25	62.63	
		P5	34.74	54.10	56.58	5.21	89.74	60.45	
		P6	23.43	44.11	41.37	1.39	79.20	50.77	
		P7	37.61	57.52	63.38	1.26	69.47	50.84	
Large Models	Qwen2.5-72B-Instruct	P1	25.07	44.89	44.73	1.42	78.14	50.82	50.44
		P2	27.09	43.12	52.09	1.56	69.72	47.40	
		P3	37.37	58.23	59.84	3.65	89.16	60.85	
		P4	36.66	61.05	61.22	8.03	89.60	62.30	
		P5	32.66	52.24	55.39	3.46	81.47	55.46	
		P6	22.19	41.17	43.27	0.47	73.14	47.33	
		P7	17.84	25.75	29.49	1.72	42.49	28.90	
	Llama-2-70B-chat	P1	25.86	39.03	44.35	0.36	63.31	42.65	48.51
		P2	23.40	37.29	39.65	0.05	64.34	42.21	
		P3	29.01	56.45	49.71	6.46	84.93	57.27	
		P4	24.77	46.48	42.10	3.87	78.03	51.12	
		P5	26.07	42.30	44.71	2.14	73.91	48.69	
		P6	24.74	45.40	41.01	1.37	76.56	49.67	
		P7	26.08	47.96	42.29	4.00	70.99	47.93	
Ultra-Large Models	Mixtral-8x22B-Instruct-v0.1	P1	26.65	45.88	50.53	0.69	78.07	51.48	49.16
		P2	14.30	28.67	28.97	0.56	57.02	35.82	
		P3	39.75	61.54	62.38	8.69	90.16	63.19	
		P4	35.14	57.33	57.31	7.68	86.87	59.95	
		P5	17.01	28.99	33.15	0.14	59.45	37.67	
		P6	23.53	45.95	42.40	0.63	79.80	51.21	
		P7	20.36	34.59	39.04	1.24	70.37	44.83	
	Llama-4-Scout-17B-16E-Instruct 109B	P1	21.78	36.89	40.36	0.08	67.38	43.61	48.93
		P2	26.63	46.20	51.81	1.40	80.58	53.03	
		P3	35.15	52.04	52.90	13.17	76.67	54.98	
		P4	36.76	59.24	59.72	5.86	88.50	60.99	
		P5	23.52	37.95	38.58	4.49	59.47	40.64	
		P6	13.29	27.71	27.29	0.85	51.98	32.99	
		P7	34.07	54.91	55.65	4.30	81.94	56.29	